

7. GOVERNANCE & RELEASE GATE · STPA Hazard Mitigation · Bumpless Override · Defensible Release Case**1. SENSING**

Photons / Voltages to DMA

MIPI CSI-2 · SPI Bus Priority

2. PERCEPTION

ViT Encoders · DINOv2

3D Spatial Affordance Tokens

3. MEMORYTemporal Belief · $SE(3)$ Trees

JEPA / RSSM World Models

5. PLANNING (System 1.5 · MPU/NPU)

Diffusion Policies · ACT Action Chunking

 H -Step Action Horizons · Jerk Continuity**4. REASONING (System 2 · MPU)**

Vision-Language Foundation Models · 3D Goals

Leases with Expiring TTL (t_{expire})**THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY****6. REFLEX (System 1 · MCU)**1 kHz Real-Time Loop · Control Barrier Functions ($h(\mathbf{x}) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$ **THE PHYSICAL WORLD** ($W_t \rightarrow W_{t+1}$) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics